

Phase 2 Implementation Plan

Telecom Diode Laser Simulator Development

1. Executive Summary & Design Philosophy

This plan outlines Phase 2 of the development of a fully coupled, self-consistent telecom diode laser simulator. Our design keeps the coupling of CarriersFVMSolver (drift-diffusion), PhotonSolver (photon density transport), and ThermoSolver (lattice heat equation) largely in their current status, securing numerical stability. Rather than projecting carrier recombination rates onto normalized modes inside the carrier loop, we implement a decoupled 2D-1D multiscale coupling paradigm. EMSolver solves the 2D transverse Helmholtz eigenvalue problem and the 1D longitudinal intensity equations. The longitudinal escape rate of photons is fed back dynamically to tune the cavity loss of the 2D PhotonSolver.

2. 2D-1D Multiscale Solver Coupling

- Step 1: EMSolver solves the 2D transverse Helmholtz eigenvalue problem to find the normalized waveguide mode profile $E_{2d}(y,z)$ and the optical confinement factor Γ .
- Step 2: The local material gain $g(y,z,n,p,T)$ is integrated across the MQW active region weighted by E_{2d} to compute the modal gain G_m .
- Step 3: The 1D Longitudinal Solver solves the traveling wave equations along the propagation axis (x) to find the longitudinal photon intensity profile $P(x)$ and the escape power P_{escape} .
- Step 4: The ratio of escape power to the midpoint intensity is computed: $\alpha_{\text{escape}} = P_{\text{escape}} / (P_{\text{mid}} * L_{\text{cav}})$.
- Step 5: The effective escape coefficient is passed back to tune the decay/loss rate of the 2D PhotonSolver: $\alpha_{\text{decay}} = v_g * \alpha_i + v_g * \alpha_{\text{escape}}$.

3. PhotonSolver PDE & Recombination Rates

PhotonSolver solves the 2D diffusion-reaction equation for the real scalar photon density $s_{\text{ph}}(y,z)$:

$$-\text{div}(D_{\text{ph}} * \text{grad}(s_{\text{ph}})) + \alpha_{\text{decay}} * s_{\text{ph}} = R_{\text{source}}$$

Where:

- D_{ph} is the photon diffusivity.
- R_{source} is the local optical generation rate: $R_{\text{source}} = \beta_{\text{sp}} * R_{\text{sp}} + R_{\text{stim}}$
- $R_{\text{sp}} = B * (n * p - n_i^2)$ is the local spontaneous recombination rate.
- $R_{\text{stim}} = v_g * g(y,z,n,p,T) * s_{\text{ph}}(y,z)$ is the local stimulated recombination rate.

This maintains physical consistency: stimulated emission occurs only where the optical field overlaps with the gain medium, preventing non-physical carrier depletion.

4. Material Perturbations & Derivatives

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The real part of the refractive index is a spatially varying profile: $n_r(y,z) = n_{\text{bulk}} + dn_T + dn_C$.

1. Temperature Dependence (Thermo-Optic Effect):

$dn_r / dT = 2.0e-4 / K$ (positive derivative, causing thermal lensing/mode confinement).

2. Carrier Dependence (Plasma Dispersion & Band-Filling):

$dn_r / dn = -2.5e-20 \text{ cm}^3$ (negative derivative, causing index reduction in the active region).

These gradients are fed back to the transverse Helmholtz solver at each iteration to re-compute $E_{2d}(y,z)$.

Status: Approved for Development (Phase 2)

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